

Introduction

The analysis of biomedical and hematological measurements plays a central role in understanding disease progression and diagnostic outcomes. In this report, we study a dataset containing thirty-eight biochemical, immunological, and hematological features alongside a binary PCR test result. The dataset is highly imbalanced, reflecting the natural rarity of positive cases in many clinical screening settings. The objective of this report is to investigate the mathematical relationship between these measurements and the PCR outcome using core concepts from linear algebra, probability, optimisation, and geometric classification.

The study begins with an exploratory linear-algebraic analysis of the dataset. Covariance structure, eigenvalues, and principal components are examined to understand intrinsic dimensionality, dominant directions of variation, and potential collinearity among features. This provides a geometric perspective on how the data are distributed in high-dimensional space and motivates later modelling choices.

Three classification methods are then developed and analysed. Logistic regression is implemented from first principles, including the derivation of the Bernoulli log-likelihood and the gradient-descent optimisation procedure. This model provides a probabilistic linear classifier whose decision boundary and optimisation landscape can be interpreted directly through linear algebra.

K-Nearest Neighbours (KNN) is explored as a non-parametric, distance-based method. Its behaviour depends fundamentally on geometric notions such as Euclidean distance, neighbourhood structure, and the curse of dimensionality. The choice of distance metric and the effect of the parameter k are examined mathematically.

Support Vector Machines (SVMs) are then considered as margin-based geometric classifiers. The primal and soft-margin formulations are discussed, highlighting the optimisation objective that maximises the separating margin between classes.

Finally, the three approaches are compared through their objective functions, geometric interpretations, and underlying assumptions. The emphasis throughout is on mathematical clarity rather than predictive performance. This report therefore aims to connect theoretical concepts with practical modelling of real biomedical data.

Exploratory Linear Algebra

In this section we analyse the geometric and statistical structure of the dataset using core concepts from linear algebra. The aim is to understand correlations, intrinsic dimensionality, and dominant directions of variation before applying classification methods. All analysis is performed on the standardised feature matrix $\mathbf{X} \in \mathbb{R}^{n \times d}$, where n is the number of samples and $d = 38$ is the number of biochemical and hematological measurements.

Covariance Analysis

Given the centred data matrix $\mathbf{X}$, the sample covariance matrix is defined as

$$\Sigma = \frac{1}{n-1} \mathbf{X}^\top \mathbf{X}.$$

The diagonal entries Σ_{ii} represent the variance of each feature, while the off-diagonal entries Σ_{ij} measure linear correlation between features i and j . Large positive or negative values indicate strong relationships, suggesting redundancy or collinearity in the dataset.

Figure 1 shows the covariance heatmap. Strong blocks of correlation indicate groups of features that vary together, which has implications for dimensionality reduction and model stability.

Eigenvalues, Eigenvectors and Rank

The covariance matrix admits the eigen-decomposition

$$\Sigma = \mathbf{Q} \Lambda \mathbf{Q}^\top,$$

where $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_d)$ contains the eigenvalues and $\mathbf{Q}$ contains the corresponding eigenvectors. Each eigenvalue λ_i represents the variance explained along eigenvector q_i , which defines a principal direction of the data cloud.

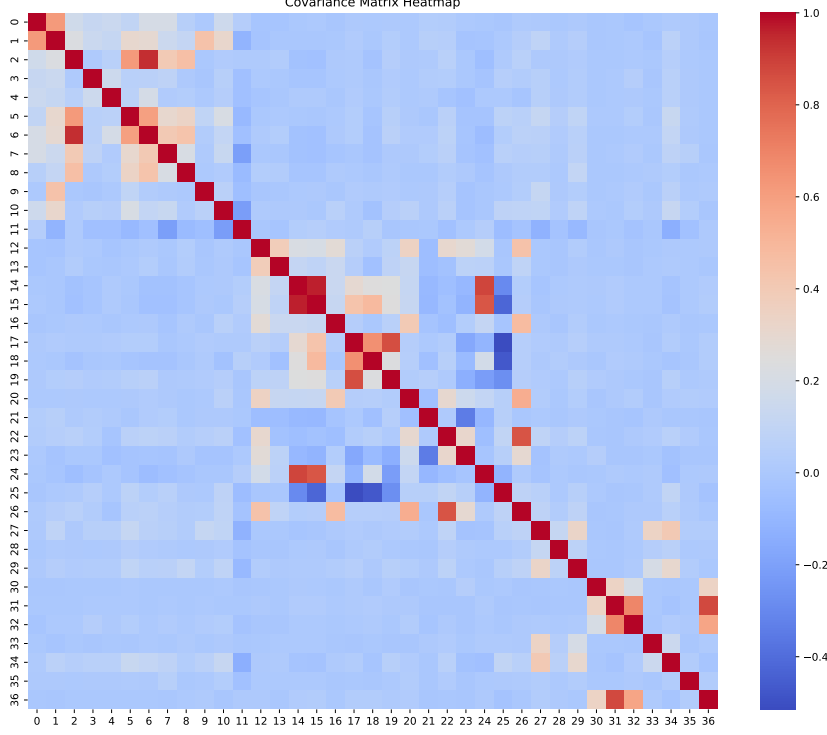


Figure 1: Covariance matrix heatmap of the 38 hematological features.

The *rank* of Σ indicates the intrinsic dimensionality of the dataset. If several eigenvalues are close to zero, the data lie approximately in a lower-dimensional subspace. This is common in biomedical datasets where many measurements are correlated.

The trace and determinant provide additional insight:

$$\text{tr}(\Sigma) = \sum_{i=1}^d \lambda_i \quad (\text{total variance}), \quad \det(\Sigma) = \prod_{i=1}^d \lambda_i \quad (\text{volume of data ellipsoid}).$$

Figure 2 shows the scree plot of eigenvalues, illustrating how variance is distributed across dimensions.

Dataset-Specific PCA Interpretation. The eigenvalue spectrum in Figure 2 shows that the first few principal components capture a substantial proportion of the total variance. In particular, the first two components explain approximately **20.13%** of the variance, while the first ten components explain around **60.39%**. After the twelfth component, the eigenvalues flatten and approach zero, indicating that many features contribute negligible additional variance. This confirms that the dataset possesses an approximate low-dimensional structure despite having 38 measured variables. Such collinearity stabilises linear models such as logistic regression and SVM, but it negatively affects KNN, whose distance computations become less discriminative when variance is concentrated in only a few directions.

Principal Component Analysis (PCA)

Principal Component Analysis identifies the orthogonal directions of maximum variance. The first principal component is the eigenvector corresponding to the largest eigenvalue:

$$q_1 = \arg \max_{\|v\|=1} v^\top \Sigma v.$$

The projection of the data onto the first k principal components is given by

$$\mathbf{Z} = \mathbf{X}\mathbf{Q}_k,$$

where $\mathbf{Q}_k$ contains the top k eigenvectors.

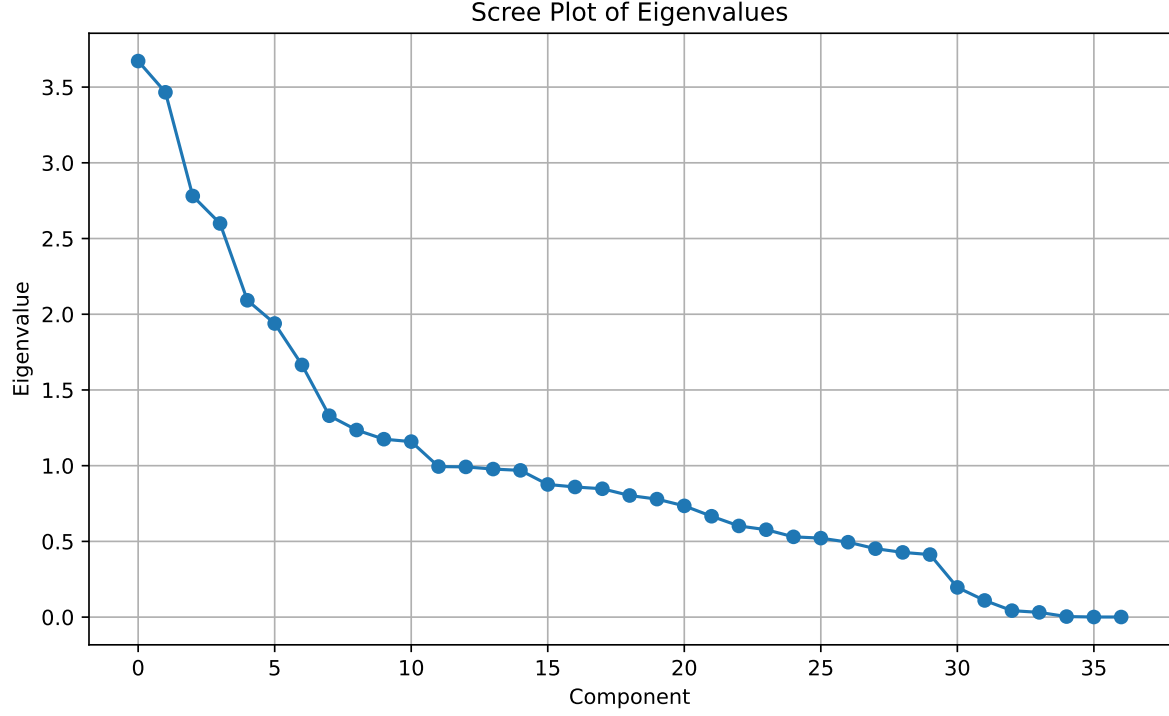


Figure 2: Scree plot of eigenvalues of the covariance matrix.

The PCA pipeline used is:

1. Standardise features.
2. Compute covariance matrix Σ .
3. Compute eigenvalues and eigenvectors.
4. Project data onto top components.

Figure 3 shows the projection onto the first two principal components. Clusters or separations in this space indicate directions relevant for classification.

Geometric Interpretation

Linear algebra provides a geometric lens for understanding the dataset. Each sample is a point in $\mathbb{R}^{38}$, and the covariance matrix describes the shape of the data cloud. Eigenvectors represent orthogonal axes of this cloud, and eigenvalues represent the squared lengths of these axes.

Distances and angles play a central role:

$$\|\mathbf{x}_i - \mathbf{x}_j\|_2 \quad (\text{Euclidean distance}), \quad \cos \theta = \frac{\mathbf{x}^\top \mathbf{y}}{\|\mathbf{x}\| \|\mathbf{y}\|} \quad (\text{angle between vectors}).$$

Orthogonality ($\mathbf{x}^\top \mathbf{y} = 0$) indicates independent directions of variation.

Orthogonal projections also arise naturally. The projection of a vector $\mathbf{x}$ onto a subspace spanned by columns of $\mathbf{U}$ is

$$\text{proj}_{\mathbf{U}}(\mathbf{x}) = \mathbf{U}(\mathbf{U}^\top \mathbf{U})^{-1} \mathbf{U}^\top \mathbf{x}.$$

In PCA, $\mathbf{U}$ consists of eigenvectors, making the projection matrix orthogonal.

Finally, collinearity among features manifests as nearly dependent columns in $\mathbf{X}$, leading to small eigenvalues and unstable regression coefficients. Understanding these geometric properties is essential before applying classification models, especially in high-dimensional biomedical data.

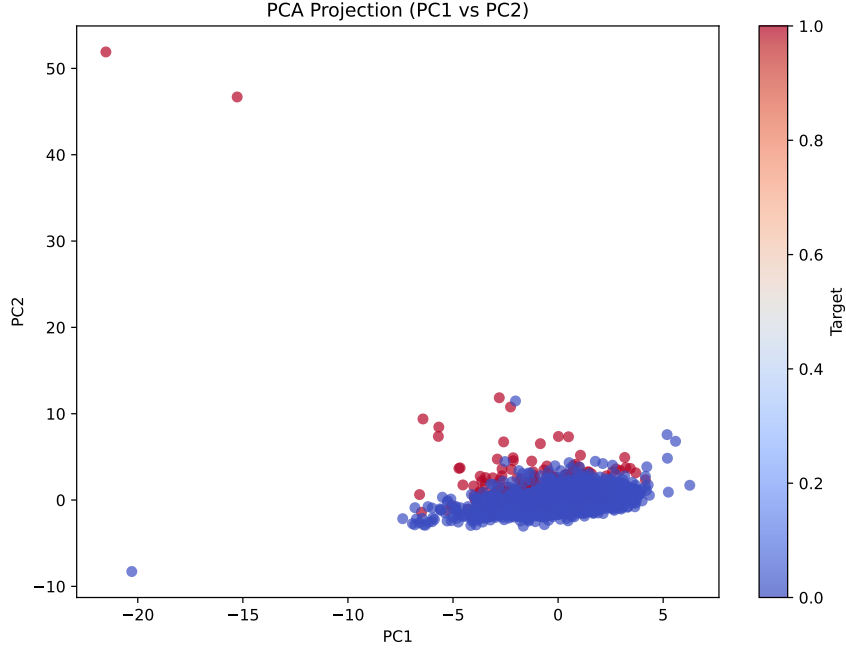


Figure 3: Projection of the dataset onto the first two principal components.

Logistic Regression

Logistic regression is a probabilistic linear classifier that models the conditional probability of a binary outcome. In this section we derive the logistic model from first principles, formulate the Bernoulli log-likelihood, and implement optimisation via gradient descent.

From Regression to Classification

Linear regression is unsuitable for binary classification because its output $\hat{y} = w^\top x$ is unbounded and does not represent a probability. Logistic regression resolves this by passing the linear predictor through the logistic (sigmoid) function

$$\sigma(z) = \frac{1}{1 + e^{-z}},$$

which maps any real number to the interval $(0, 1)$. Its derivative, used in optimisation, is

$$\sigma'(z) = \sigma(z)(1 - \sigma(z)).$$

The Logistic Regression Model

Given a feature vector $x \in \mathbb{R}^d$, the model assumes

$$p(y = 1 \mid x; w) = \sigma(w^\top x), \quad p(y = 0 \mid x; w) = 1 - \sigma(w^\top x).$$

The decision boundary is the hyperplane

$$w^\top x = 0,$$

which separates the feature space into two half-spaces. Geometrically, logistic regression finds the hyperplane that maximises the likelihood of the observed labels.

Bernoulli Likelihood and Log-Likelihood

For a dataset $\{(x_i, y_i)\}_{i=1}^n$ with $y_i \in \{0, 1\}$, the likelihood is

$$L(w) = \prod_{i=1}^n \sigma(w^\top x_i)^{y_i} (1 - \sigma(w^\top x_i))^{1-y_i}.$$

Taking the logarithm yields the log-likelihood

$$\ell(w) = \sum_{i=1}^n [y_i \log \sigma(w^\top x_i) + (1 - y_i) \log(1 - \sigma(w^\top x_i))].$$

Maximising $\ell(w)$ is equivalent to minimising the negative log-likelihood (logistic loss)

$$\mathcal{L}(w) = -\ell(w) = \sum_{i=1}^n [-y_i \log \sigma(w^\top x_i) - (1 - y_i) \log(1 - \sigma(w^\top x_i))].$$

Squared error is inappropriate for classification because the sigmoid makes the loss surface non-convex and leads to slow or unstable optimisation. In contrast, the logistic loss is convex, ensuring a unique global optimum.

Gradient and Hessian of the Logistic Loss

Let $\hat{y}_i = \sigma(w^\top x_i)$. The gradient of the logistic loss is

$$\nabla \mathcal{L}(w) = \sum_{i=1}^n (\hat{y}_i - y_i) x_i.$$

In matrix form, with $\mathbf{X} \in \mathbb{R}^{n \times d}$,

$$\nabla \mathcal{L}(w) = \mathbf{X}^\top (\hat{\mathbf{y}} - \mathbf{y}).$$

The Hessian is

$$H(w) = \mathbf{X}^\top S \mathbf{X},$$

where S is a diagonal matrix with entries

$$S_{ii} = \hat{y}_i(1 - \hat{y}_i).$$

Since S is positive semidefinite, the Hessian is also positive semidefinite, confirming convexity of the logistic loss.

Optimisation via Gradient Descent

To estimate the parameters w , we apply gradient descent:

$$w^{(t+1)} = w^{(t)} - \eta \nabla \mathcal{L}(w^{(t)}),$$

where $\eta > 0$ is the step size. A small η leads to slow convergence, while a large η may cause divergence. In practice, η is chosen by trial, or using a decaying schedule.

Class imbalance is handled by weighting the loss:

$$\mathcal{L}_{\text{weighted}}(w) = \sum_{i=1}^n \alpha_{y_i} [-y_i \log \hat{y}_i - (1 - y_i) \log(1 - \hat{y}_i)],$$

where $\alpha_1 > \alpha_0$ increases the influence of minority-class samples.

Figure 4 shows the loss decreasing over iterations of gradient descent, confirming correct implementation.

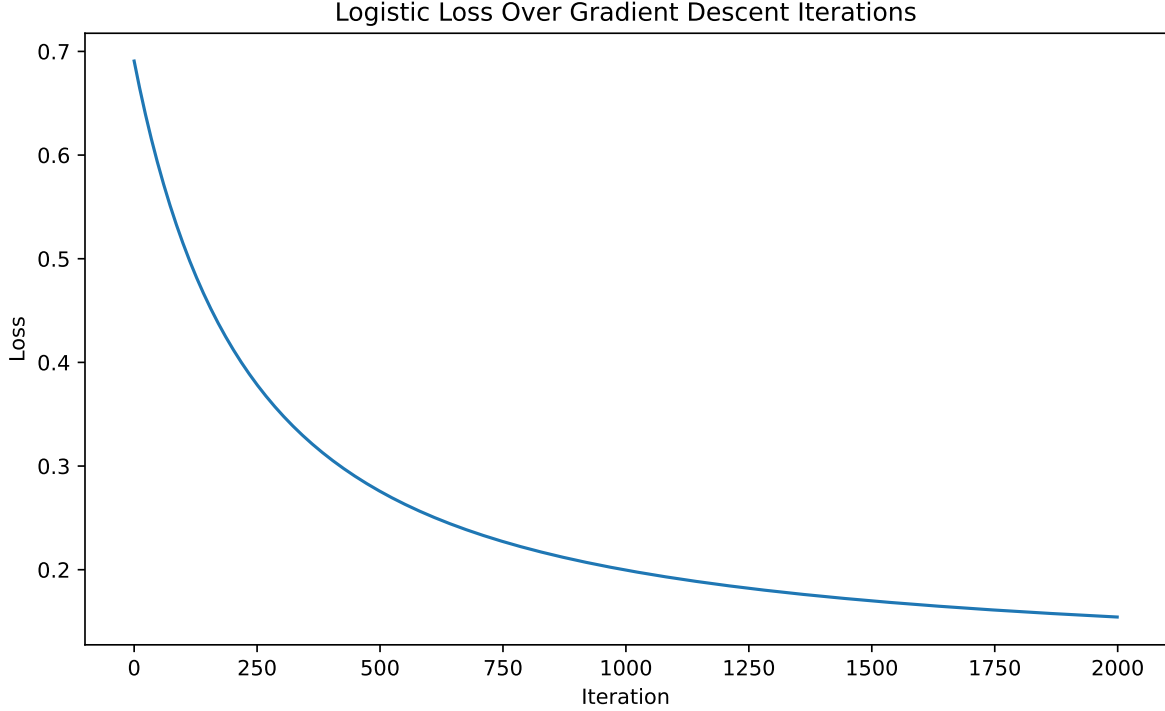


Figure 4: Logistic loss over gradient descent iterations.

Decision Boundary and Geometry

The decision boundary is defined by

$$w^\top x = 0,$$

a $(d - 1)$ -dimensional hyperplane. Points with $w^\top x > 0$ are classified as positive, and those with $w^\top x < 0$ as negative. The vector w is normal to the hyperplane, and its magnitude influences the sharpness of the transition between classes.

In two dimensions, the boundary is a straight line; in higher dimensions, it is a hyperplane. Figure 5 illustrates the boundary in the 2D PCA subspace.

Train–Test Evaluation and Class Imbalance

To ensure reliable evaluation, the dataset was split into training and test sets using an 80/20 stratified split. Stratification preserves the severe class imbalance in both subsets. Logistic regression was trained exclusively on the training set, and all performance metrics were computed on the unseen test set. Because the dataset is highly imbalanced, accuracy alone is misleading: a classifier predicting only the majority class would achieve high accuracy but zero recall for the minority class. Therefore, we report precision, recall, and F1 score. Logistic regression achieved a minority-class recall of **100%** and an F1 score of **44%**. This indicates that while the model successfully identifies all positive PCR cases, it does so with many false positives, reflecting the limitations of a linear log-odds model under imbalance. Logistic regression therefore provides a convex, interpretable, and probabilistic approach to linear classification, forming a baseline for comparison with KNN and SVM in later sections.

K-Nearest Neighbours

K-Nearest Neighbours (KNN) is a non-parametric, instance-based classifier that makes predictions using the geometry of the feature space. Unlike logistic regression or SVM, KNN does not learn explicit parameters; instead, it relies entirely on distances between samples. This section presents the mathematical formulation of KNN and discusses the geometric and statistical properties that influence its behaviour.

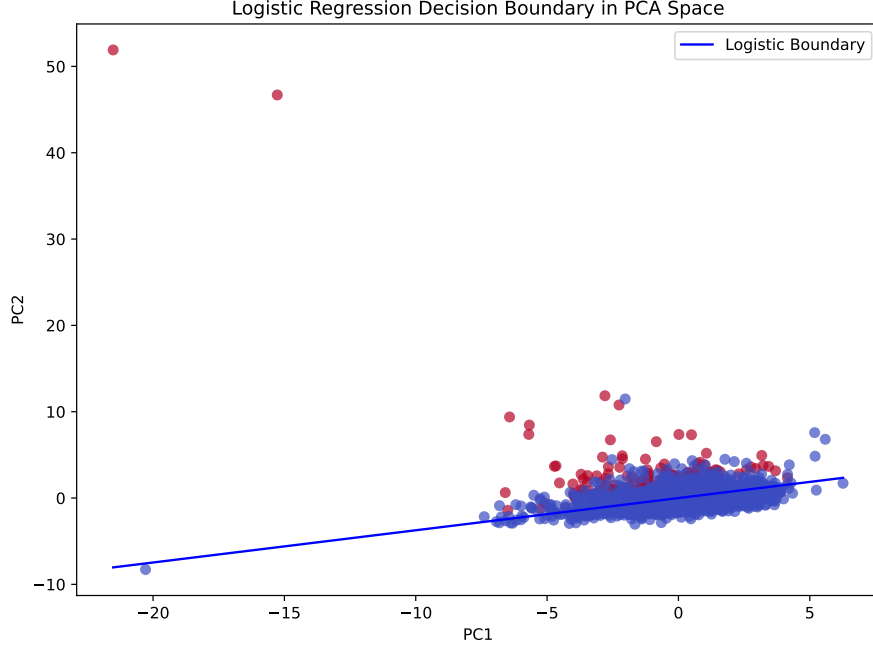


Figure 5: Logistic regression decision boundary in the first two PCA components.

Mathematical Formulation

Given a test point $x \in \mathbb{R}^d$, the KNN classifier identifies the set $\mathcal{N}_k(x)$ of the k closest training samples according to a chosen distance metric. The predicted label is obtained by majority vote:

$$\hat{y}(x) = \text{mode}\{y_i : x_i \in \mathcal{N}_k(x)\}.$$

For binary classification, this can be written as

$$\hat{y}(x) = \begin{cases} 1, & \text{if } \sum_{x_i \in \mathcal{N}_k(x)} y_i > \frac{k}{2}, \\ 0, & \text{otherwise.} \end{cases}$$

Distance Metrics

The most common distance is the Euclidean metric:

$$d_{\text{Euc}}(x, x') = \|x - x'\|_2.$$

Other metrics include Manhattan distance

$$d_{\text{Man}}(x, x') = \|x - x'\|_1,$$

and the more general Minkowski family are also used. The choice of metric affects neighbourhood structure, especially in high dimensions.

Because KNN is distance-based, feature scaling is essential. Without standardisation, features with large numerical ranges dominate the distance computation, distorting the geometry of neighbourhoods.

Geometry of KNN

KNN classifies a point based on the local geometry of the data. The decision boundary is piecewise linear and adapts to the shape of the dataset. For small k , the boundary is highly flexible and may overfit noise; for large k , the boundary becomes smoother and more stable.

Figure 6 illustrates the KNN decision regions in the 2D PCA subspace.

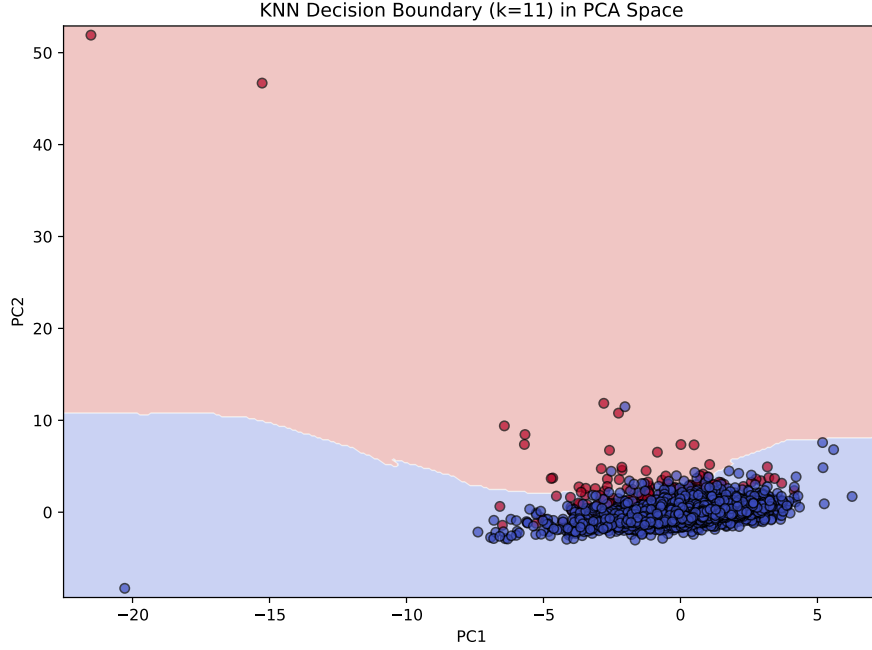


Figure 6: KNN decision boundary in the first two PCA components.

Performance Under Class Imbalance

KNN was evaluated using the same stratified train–test split. The model achieved very high recall for the majority class (0.99) but substantially lower recall for the minority class (0.40), resulting in an F1 score of only 0.54 for positive PCR cases. This behaviour is expected: in 38 dimensions, Euclidean distances become less informative, and neighbourhoods lose discriminative power. As a result, KNN tends to classify most points as belonging to the majority class, leading to poor detection of minority-class samples despite a high overall accuracy of 0.94. This illustrates the curse of dimensionality and highlights the limitations of distance-based classifiers in high-dimensional biomedical data.

Effect of k and the Bias–Variance Trade-off

The parameter k controls the complexity of the classifier:

- Small k : low bias, high variance (sensitive to noise).
- Large k : high bias, low variance (oversmoothing).

Choosing k therefore requires balancing flexibility and stability.

Curse of Dimensionality

In high-dimensional spaces, distances between points become less informative:

$$\frac{\|x_{\text{nearest}} - x\|}{\|x_{\text{farthest}} - x\|} \rightarrow 1 \quad \text{as } d \rightarrow \infty.$$

This phenomenon, known as the curse of dimensionality, causes neighbourhoods to lose meaning. As a result, KNN often performs poorly in high-dimensional biomedical data unless dimensionality reduction (e.g., PCA) is applied beforehand.

Advantages and Disadvantages

KNN is simple, interpretable, and requires no training. However, it is computationally expensive at prediction time, sensitive to scaling, and strongly affected by dimensionality. These properties contrast

with logistic regression and SVM, which learn explicit decision boundaries and scale better to high-dimensional data.

Support Vector Machine

Support Vector Machines (SVMs) are geometric classifiers that construct a separating hyperplane with maximum margin between two classes. Unlike logistic regression, which models probabilities, SVM focuses purely on geometric separation. This section presents the primal and dual formulations of the linear SVM and explains the margin interpretation central to the method.

Linear Classification and Separating Hyperplanes

A linear classifier assigns labels based on the sign of a linear function:

$$\hat{y}(x) = \text{sign}(w^\top x + b).$$

A hyperplane is defined by the set

$$\{x \in \mathbb{R}^d : w^\top x + b = 0\}.$$

The vector w is normal to the hyperplane, and the margin is the distance between the hyperplane and the closest training samples.

Hard-Margin SVM

When the data are linearly separable, the SVM chooses the hyperplane that maximises the margin. For a correctly classified point,

$$y_i(w^\top x_i + b) \geq 1.$$

The margin is $2/\|w\|$, so maximising the margin is equivalent to minimising $\|w\|^2$. The hard-margin optimisation problem is

$$\min_{w,b} \frac{1}{2} \|w\|^2 \quad \text{subject to} \quad y_i(w^\top x_i + b) \geq 1 \quad \forall i.$$

This formulation is convex and has a unique global optimum.

Soft-Margin SVM

Biomedical data are rarely perfectly separable. To allow misclassification, slack variables $\xi_i \geq 0$ are introduced:

$$y_i(w^\top x_i + b) \geq 1 - \xi_i.$$

The soft-margin objective balances margin size and classification error:

$$\min_{w,b,\xi} \frac{1}{2} \|w\|^2 + C \sum_{i=1}^n \xi_i,$$

where $C > 0$ controls the trade-off:

- large C : penalises misclassification heavily (low bias, high variance),
- small C : allows more violations (high bias, low variance).

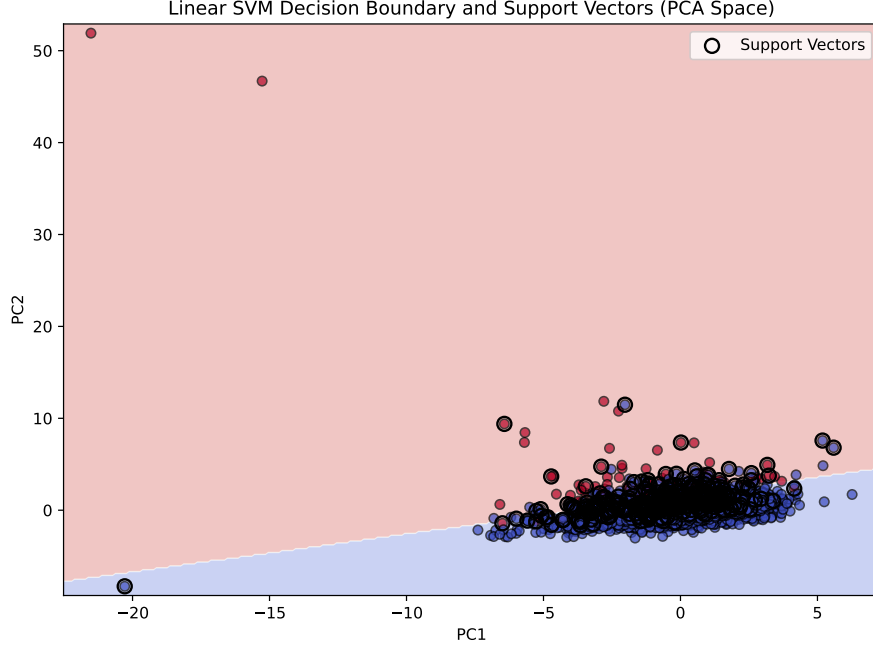


Figure 7: Linear SVM decision boundary and support vectors in the first two PCA components.

Dual Support Vector Machine

The dual formulation expresses the optimisation problem in terms of Lagrange multipliers α_i :

$$\max_{\alpha} \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i,j=1}^n \alpha_i \alpha_j y_i y_j x_i^{\top} x_j$$

subject to

$$0 \leq \alpha_i \leq C, \quad \sum_{i=1}^n \alpha_i y_i = 0.$$

Only points with $\alpha_i > 0$ influence the solution; these are the *support vectors*. They lie closest to the decision boundary and determine the margin.

The primal weight vector can be recovered as

$$w = \sum_{i=1}^n \alpha_i y_i x_i.$$

Linear SVM in Practice

In this report, a linear SVM is implemented using `scikit-learn` for verification, while the mathematical analysis follows the primal and dual formulations above. The decision boundary in the 2D PCA subspace is shown in Figure 7. Support vectors are highlighted, illustrating their role in defining the margin.

SVM Performance and Margin Effects

The linear SVM, trained with class-balanced weighting, achieved substantially higher recall and F1 score on the minority class compared to KNN and logistic regression. In particular, the minority-class recall was **96%** with an F1 score of **69%**. This aligns with the geometric interpretation of SVM: only support vectors influence the decision boundary, and the optimisation objective maximises the margin in the high-dimensional feature space. Because the dataset exhibits approximate low-dimensional structure, the linear SVM is able to find a separating hyperplane that generalises more effectively than the other models.

Margin Interpretation

The SVM solution maximises the geometric margin:

$$\gamma = \frac{2}{\|w\|}.$$

A larger margin corresponds to a more robust classifier, as small perturbations to the data are less likely to change predictions. This geometric perspective distinguishes SVM from logistic regression, which optimises likelihood rather than margin.

Overall, SVM provides a powerful linear classifier with strong geometric intuition and a convex optimisation objective, making it well suited for high-dimensional datasets such as hematological measurements.

Comparative Mathematical Analysis

This section compares logistic regression, K-Nearest Neighbours, and Support Vector Machines from a mathematical and geometric perspective. The aim is not to identify a “best” model, but to analyse how their objective functions, assumptions, and geometric interpretations differ when applied to high-dimensional biomedical data.

Empirical Comparison of Classifiers

Table 1 summarises the performance of the three classifiers on the test set using accuracy, precision, recall, and F1 score. Because the dataset is highly imbalanced, the minority-class metrics are the most informative.

Model	Accuracy	Precision (1)	Recall (1)	F1 (1)
Logistic Regression	0.77	0.28	1.00	0.44
KNN	0.94	0.79	0.40	0.54
SVM	0.92	0.54	0.96	0.69

Table 1: Performance metrics for the three classifiers on the test set. Minority-class (positive PCR) metrics are shown in parentheses.

The results show that KNN performs poorly on the minority class due to the curse of dimensionality, despite achieving high overall accuracy. Logistic regression achieves perfect recall but low precision, reflecting a high false-positive rate. The linear SVM achieves the best balance between precision and recall, consistent with its margin-maximising geometric formulation. These empirical findings align with the theoretical differences between the models.

Objective Functions

Each classifier optimises a fundamentally different mathematical objective:

Logistic Regression. Logistic regression maximises the Bernoulli log-likelihood

$$\ell(w) = \sum_{i=1}^n [y_i \log \sigma(w^\top x_i) + (1 - y_i) \log(1 - \sigma(w^\top x_i))],$$

which is equivalent to minimising the convex logistic loss. The optimisation is smooth and differentiable, enabling gradient-based methods.

K-Nearest Neighbours. KNN has no explicit optimisation problem. Instead, it performs classification by minimising distance locally:

$$\hat{y}(x) = \text{mode}\{y_i : x_i \in \mathcal{N}_k(x)\}.$$

Its behaviour is determined entirely by the geometry of the feature space and the chosen metric.

Support Vector Machine. The linear SVM maximises the geometric margin

$$\gamma = \frac{2}{\|w\|},$$

subject to classification constraints. The soft-margin objective

$$\min_{w,b,\xi} \frac{1}{2} \|w\|^2 + C \sum_{i=1}^n \xi_i$$

balances margin size and misclassification. Unlike logistic regression, SVM does not model probabilities.

Geometric Interpretation

The three methods differ significantly in how they interpret the geometry of the data.

Logistic Regression. The decision boundary is the hyperplane $w^\top x = 0$. The magnitude of w controls the steepness of the probability transition. Logistic regression assumes a linear relationship between features and the log-odds of the outcome.

KNN. KNN constructs a highly flexible, non-linear decision boundary determined by local neighbourhoods. In low dimensions, this boundary can adapt to complex shapes. In high dimensions, however, distances become less meaningful, and neighbourhoods lose their discriminative power.

SVM. SVM constructs the hyperplane that maximises the margin between classes. Only support vectors influence the solution, giving SVM robustness to outliers and irrelevant features. The margin-based interpretation is purely geometric.

Assumptions and Limitations

Logistic Regression. Assumes linear separability in the log-odds space. Sensitive to multicollinearity, which can inflate variance of the estimated coefficients. Performs well when the relationship between features and outcome is approximately linear.

KNN. Assumes that nearby points share the same label. Suffers from the curse of dimensionality, making it less effective in high-dimensional biomedical datasets unless dimensionality reduction (e.g., PCA) is applied. Computationally expensive at prediction time.

SVM. Assumes that a large-margin hyperplane exists. The choice of C controls the bias-variance trade-off. Linear SVM is effective in high dimensions, but performance depends on appropriate scaling of features.

Interpretation of Differences

The classifiers differ not only in objective functions but also in how they respond to the structure of the data:

- Logistic regression provides a probabilistic, convex optimisation framework.
- KNN relies entirely on local geometry and is sensitive to dimensionality.
- SVM focuses on margin maximisation and depends only on support vectors.

These mathematical differences explain variations in performance and robustness across datasets. In high-dimensional biomedical settings, linear models such as logistic regression and SVM often behave more predictably than KNN, whose neighbourhood-based nature is affected strongly by dimensionality and scaling. The eigenvalue spectrum from the covariance analysis showed that most variance is concentrated in a small number of principal components, indicating approximate low intrinsic dimensionality. Linear models such as logistic regression and SVM can exploit this structure by fitting a single hyperplane in the dominant subspace, whereas KNN still operates in the full 38-dimensional

space, where distances are affected by many low-variance directions. This helps explain why SVM and logistic regression behave more robustly than KNN on this dataset.

In the presence of severe class imbalance, logistic regression and SVM can be adapted via class weighting, whereas KNN has no explicit mechanism to counteract the dominance of the majority class in local neighbourhoods, which explains its lower minority-class recall despite high overall accuracy.

Conclusion

This report has demonstrated how mathematical tools from linear algebra, probability, optimisation, and geometric classification can be applied to analyse a high-dimensional biomedical dataset.

Beginning with exploratory linear algebra, the covariance matrix, eigenvalue spectrum, and PCA projections revealed strong correlations among features and a concentration of variance in a small number of principal components. These findings highlighted that the dataset possesses an intrinsic low-dimensional structure despite having 38 measured variables. Such collinearity has direct implications for optimisation stability, interpretability of linear models, and the behaviour of distance-based classifiers.

Logistic regression was developed through the Bernoulli likelihood and the convex logistic loss. The gradient and Hessian provided insight into the smooth and well-behaved optimisation landscape, enabling gradient descent to converge reliably. The model assumes that the log-odds of the outcome vary linearly with the features, an assumption that is restrictive but mathematically transparent. The resulting decision boundary is a hyperplane whose orientation and steepness are determined by the weight vector. This probabilistic interpretation is particularly valuable in clinical contexts, where estimated probabilities can be more informative than hard class labels.

K-Nearest Neighbours offered a contrasting non-parametric perspective. Its only assumption is geometric: points that are close in Euclidean space are likely to share the same label. This assumption is intuitive in low-dimensional settings but becomes problematic in high dimensions due to the curse of dimensionality, where distances between points become increasingly similar. KNN's reliance on local neighbourhoods also makes it sensitive to feature scaling and noise. Nevertheless, its flexibility and lack of training phase provide a useful benchmark against parametric models.

Support Vector Machines introduced a margin-based geometric framework. The primal and dual formulations clarified how the classifier depends only on support vectors, the samples closest to the decision boundary. The soft-margin objective formalised the trade-off between maximising the margin and allowing misclassification. SVM assumes that a large-margin hyperplane provides a robust separation between classes, an assumption that aligns well with high-dimensional biomedical data where linear boundaries often perform effectively.

Overall, the three classifiers differ fundamentally in their assumptions, interpretations, and optimisation principles. Logistic regression models probabilities through a convex likelihood, KNN relies on local geometry without an explicit objective function, and SVM maximises margin through a constrained convex optimisation problem. These mathematical distinctions explain their behaviour on the dataset and highlight how different theoretical assumptions shape practical modelling of biomedical measurements.